Memory Locker

A self-hosted extension to ChatGPT that stores and retrieves memories or files on your private database with rich context

Presentation of MVP solution (v1.0 - April 14, 2025)

As you talk to ChatGPT, rich context is appended to each interaction

> *"I ran into Stephen Barkin today at Starbucks. He mentioned he is using a platform called Quantrix for his trade analysis. He asked me to call him next week to schedule some demo time. Could be useful for Project Mintra. I should get back to him by Monday morning."*

Memory locker pulls entities from your input

People

- Stephen Barkin

Locations

- Starbucks

Organisations

- Quantrix
- Project Mintra

It converts your relative dates to indexable entities

"Today" & "Monday"

- Month DD, YYYY

"Next week"

- Week of month

"Monday morning"

- Period: morning

Before storing, it also infers context from your content

Topics

- Trade analysis
- Platform demo

Memory type

- Casual encounter
- Action item

Priority

- High

Three-layer architecture orchestrates and secures your interactions

Front end GPT assistant

You interact with ChatGPT via the official app or website

Makes intelligent decisions about how to process your natural language inputs (voice, text, file upload)

Middleware memo processor

Works in the background to process and organise your memories

Takes structured contextual output from GPT and tags it with rich metadata

Your private database

Where your information is stored securely

Your complete input plus rich metadata is indexed for quick, precise, retrieval later

Breaking down the flow: Storing a memory

Front end GPT assistant

You provide information in natural language (i.e. any way you please)

GPT identifies key elements and infers entities that will add context to your stored memo

All input is translated to English and output in a structured, predictable, schema.

Middleware memo processor

A self-hosted application organises the information into structured metadata, normalises dates and times, and otherwise prepares your content for database store

Private memory vault

Stores your memo in a hybrid database:

1. Vector store (what GPT uses to quickly identify related contexts)

2. Postgres filing system (what classic databases use to retrieve precise file information)

Breaking down the flow: Retrieving a memory

Front end GPT assistant

GPT understands whether your intent is to store or query your data, and prepares the request accordingly

* If your interaction includes both a store and query, it will process each sequentially.

Middleware memo processor

Analyses your query

Matches query keywords against stored metadata (people, dates, topics, etc…)

Ranks results by relevance to your question

Considers time periods and context

Private memory vault

Memory Vault retrieves the best matches

Combines different types of matches for better results (vector similarity search + file text search)

Returns matches to middleware for ranking best matches

Feeds back context from connected memories that boost ranking

Development, documentation, & setup

The Memory Locker is a sophisticated AI-powered system that helps you store and retrieve information using natural language. Its three-layer architecture work together to understand, organise, and store your memories.

The system uses advanced processing to identify key elements like people, places, dates, and topics, making it easy to find exactly what you're looking for when you need it.

Next features of development will include adding a separate function for editing stored memories by interacting with ChatGPT.

Project source files, database schema, and customGPT action schema can be found on a github repo. Setup instructions are included but will require knowledge of how to implement web-hosted functions.

Github repository:

https://github.com/Choice-Architect/memory-locker

https://github.com/Choice-Architect/memory-locker/blob/main/README.md